Weak-noise scaling of the quantum reservoir memory bound

A proposed matching upper bound and classical construction

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Abstract

Consider finite quantum reservoirs with local depolarizing noise of retention parameter $0 < q < 1$, k input qubits reset at each time step, and arbitrary fixed reservoir unitaries. This note develops a refinement of the memory ceiling in Ref. [1]. A block argument based on reverse hypercontractivity improves the upper exponent from $O(k/(1-q)^2)$ to $O(k\log(1/(1-q))/(1-q))$ in the weak-noise limit. A family of classical rank-compression reservoirs attains the latter scaling using $k \geq 4$ reset qubits. The resulting proposed law is

$$\log W_{q,k}(\varepsilon) = \Theta\left(\frac{k}{1-q}\log\frac{1}{1-q}\right) \quad (q \uparrow 1,\ \varepsilon \text{ fixed},\ k \geq 4).$$

The proof includes a finite-dimensional implementation, a stationary-limit argument, and input histories witnessing echo-state memory. In particular, a bound $\exp[O(k/(1-q))]$ with a constant uniform in q is incompatible with this construction.

1 Model and precise result

At each discrete time, k input qubits are reset to an input-dependent state $\sigma(u)$, a fixed unitary U acts on all n qubits, and every qubit receives depolarizing noise:

$$\Phi_u(\rho) = \mathcal{D}_q^{\otimes n}\left(U[\sigma(u)\otimes \mathrm{Tr}_{\mathrm{in}}\,\rho]U^\dagger\right), \qquad \mathcal{D}_q(A) = qA + (1-q)\,\mathrm{Tr}(A)\frac{I}{2}. \tag{1.1}$$

No restriction is placed on n, the encoding, or the complexity of U. The upper proof also permits arbitrary, time-dependent CPTP maps E_t with

$$E_t(I) \leq 2^k I, \qquad \Phi_t = \mathcal{N}_q E_t, \qquad \mathcal{N}_q = \mathcal{D}_q^{\otimes n}. \tag{1.2}$$

Thus it applies to the entropy-budget formulation of Ref. [1].

Let $\rho(h)$ be the echo state associated with an infinite input history. For a given reservoir, $W(\varepsilon)$ is the smallest nonnegative integer L such that every two histories agreeing on their last L inputs obey $D_{\mathrm{tr}}(\rho(h),\rho(h')) \leq \varepsilon$, where $D_{\mathrm{tr}}(\rho,\rho') = \frac{1}{2}\|\rho-\rho'\|_1$. Write $W_{q,k}(\varepsilon)$ for the supremum over finite reservoir sizes, unitaries, and encodings. The supremum formulation does not require that an optimal finite size exist.

All logarithms without a subscript are natural. Define

$$r = q^2, \qquad \eta = 1-r, \qquad \Lambda(q) = \sum_{j=1}^{\infty}\frac{r^j}{1-r^j}, \qquad m = \left\lceil \frac{\log 2}{-\log r}\right\rceil. \tag{1.3}$$

The series is finite for every $q < 1$, and $m \geq 1$.

Theorem 1.1 (Explicit bounds)**.** *For $0 < q < 1$, $0 < \varepsilon < 1$, and every integer $k \geq 4$,*

$$W_{q,k}(\varepsilon) \leq m \left\lceil 2^{\,10+6k\Lambda(q)} \log \frac{1}{\varepsilon} \right\rceil, \tag{1.4}$$

$$W_{q,k}(\varepsilon) \geq \left\lfloor \frac{1-\varepsilon}{4\sqrt{1-q^2}} \exp\!\left(\frac{k-3}{8} \Lambda(q) \right) \right\rfloor. \tag{1.5}$$

The upper bound also holds for $k \geq 1$ and for the general channels in (1.2). The lower bound is realized by a finite classical reservoir of the form (1.1), with a fixed permutation unitary. Its required size may depend on q, k, ε.

Lemma 1.2 (Size of the series)**.** *With $\lambda = -\log(q^2)$,*

$$\frac{\log(1/\eta)}{\lambda} \leq \Lambda(q) \leq \frac{q^2}{\eta} + \frac{\log(1/\eta)}{\lambda}. \tag{1.6}$$

Consequently,

$$\Lambda(q) \sim \frac{1}{2(1-q)} \log \frac{1}{1-q} \qquad (q \uparrow 1). \tag{1.7}$$

Proof. The function $f(x) = (e^{\lambda x} - 1)^{-1}$ decreases. Comparing the sum with its integral gives $\int_1^\infty f(x)\,dx \leq \sum_{j \geq 1} f(j) \leq f(1) + \int_1^\infty f(x)\,dx$. The integral equals $\lambda^{-1} \log(1/\eta)$, which proves (1.6). Since $\lambda \sim \eta \sim 2(1-q)$, the additional term q^2/η is of lower order, proving (1.7). □

Corollary 1.3 (Matching weak-noise scaling)**.** *For each fixed $0 < \varepsilon < 1$, there are positive constants $c_\varepsilon, C_\varepsilon$ and $q_\varepsilon < 1$, independent of k, such that for all integers $k \geq 4$ and $q_\varepsilon < q < 1$,*

$$c_\varepsilon \frac{k}{1-q} \log \frac{1}{1-q} \leq \log W_{q,k}(\varepsilon) \leq C_\varepsilon \frac{k}{1-q} \log \frac{1}{1-q}. \tag{1.8}$$

In particular, no upper bound $W_{q,k}(\varepsilon) \leq \exp[C_\varepsilon k/(1-q)]$ can hold with C_ε independent of q.

Proof. Use Theorem 1.1, Lemma 1.2, and $k - 3 \geq k/4$. The logarithm of the block length and the fixed-accuracy prefactors are lower-order terms, uniformly for $k \geq 4$. The floor in the lower bound can be absorbed once q is sufficiently close to one. Fixing $k = 4$ already contradicts the proposed upper exponent $C_\varepsilon k/(1-q)$. □

2 Preliminaries for the upper bound

For $d = 2^n$, set $\tau(A) = d^{-1} \operatorname{Tr} A$. We use the normalized Schatten quantities

$$|A|_p = (\tau |A|^p)^{1/p}, \qquad |H|_0 = \exp(\tau \log H) \quad (H > 0). \tag{2.1}$$

For $0 < p < 1$, the first expression is used only on positive operators when concavity is needed. These quantities need not be norms.

Lemma 2.1 (Effect of a budgeted channel)**.** *Let E be CPTP and $E(I) \leq 2^k I$. For Hermitian X,*

$$|EX|_1 \leq |X|_1, \qquad |EX|_2 \leq 2^{k/2} |X|_2. \tag{2.2}$$

For $G \geq 0$ and $0 < p < 1$,

$$|EG|_p \geq 2^{-k(1-p)/p} |G|_p. \tag{2.3}$$

Proof. The first inequality is trace-norm contractivity. Put $B = 2^k$. The CP map E/B is subunital. The Schwarz inequality yields $(EX)^2 \le BE(X^2)$; trace preservation gives the second inequality. For (2.3), operator concavity of t^p, including $0^p = 0$, gives

$$(EG)^p \ge B^{p-1} E(G^p).$$

Taking normalized traces and then the positive p-th root proves the claim. The subunital form of Jensen follows by completing the map with a zero input block to make it unital. □

We use two hypercontractive properties of the qubit depolarizing channel. Forward hypercontractivity gives

$$|\mathcal{N}_q X|_2 \le |X|_{1+q^2}\,. \tag{2.4}$$

This is the result used in Lemma S2 of Ref. [1]; see also Ref. [3]. Reverse hypercontractivity gives, for positive input,

$$|\mathcal{N}_q G|_a \ge |G|_b\,, \qquad a \le b \le 1, \qquad q^2 \le \frac{1-b}{1-a}. \tag{2.5}$$

Theorem 4 of Ref. [2] includes this range. The endpoint $a = 0$ is understood by continuity. It is legitimate here because every nonzero positive operator becomes positive definite after a depolarizing layer.

Lemma 2.2 (Purity cap and existence of echo states)**.** *Let*

$$\theta = \frac{2q^2}{1+q^2}, \qquad P_* = 2^{kq^2/(1-q^2)}. \tag{2.6}$$

For every Hermitian X,

$$|\mathcal{N}_q EX|_2 \le P_*^{1-\theta}\, |X|_1^{1-\theta}\, |X|_2^{\theta}\,. \tag{2.7}$$

Every echo state satisfies $|d\rho(h)|_2 \le P_*$.

Proof. Interpolation, (2.4), and Lemma 2.1 give

$$|\mathcal{N}_q EX|_2 \le |EX|_{1+q^2} \le |EX|_1^{1-\theta}\, |EX|_2^{\theta} \le 2^{k\theta/2}\, |X|_1^{1-\theta}\, |X|_2^{\theta}\,.$$

The identity $2^{k\theta/2} = P_*^{1-\theta}$ proves the recursion. For $X = d\rho$, its first norm is one, so

$$\log_2 |d\rho_t|_2 \le \theta \log_2 |d\rho_{t-1}|_2 + \frac{k\theta}{2}.$$

The fixed point is $\log_2 P_* = kq^2/(1-q^2)$.

For completeness, echo-state existence at each finite n follows from the decomposition of $\mathcal{N}_q$ containing global replacement by I/d with weight $(1-q)^n > 0$. Thus every step contracts trace distance by at least the uniform factor $1-(1-q)^n < 1$, whatever the input. Taking the initial time to the infinite past gives the unique echo state. Iterating the purity recursion from $|d\rho_0|_2 \le \sqrt{d}$ proves its cap. The finite-n contraction is used only for existence; the memory bound proved below is independent of n. □

3 Block contraction and the upper bound

Let $\Psi = \Phi_m \cdots \Phi_1$, with m from (1.3), and define

$$p_j = 1 - r^j \quad (0 \le j \le m), \qquad A_m = \sum_{j=1}^{m} \frac{r^j}{1-r^j}. \tag{3.1}$$

In particular $p_0 = 0$, $p_m \ge 1/2$, and $A_m \le \Lambda(q)$.

Lemma 3.1 (Geometric mean after a block)**.** *If $G \geq 0$ and $\tau G = 1$, then*

$$|\Psi G|_0 \geq 2^{-kA_m} |G|_2^{-2} . \tag{3.2}$$

Proof. Work backward from the output exponent zero. Apply reverse hypercontractivity to the last noise layer with exponents p_0, p_1, then (2.3) to E_m. Repeat with p_1, p_2, and so on. Since $r = (1-p_j)/(1-p_{j-1})$, this gives exactly

$$|\Psi G|_0 \geq \prod_{j=1}^{m} 2^{-k(1-p_j)/p_j} |G|_{p_m} = 2^{-kA_m} |G|_{p_m} .$$

Monotonicity gives $|G|_{p_m} \geq |G|_{1/2}$. Convexity of $p \mapsto \log \tau G^p$, at $1/2, 1, 2$, and $\tau G = 1$, imply $|G|_{1/2} \geq |G|_2^{-2}$. □

Lemma 3.2 (Acceptance of a large test)**.** *If $H > 0$, $\tau H = 1$, and $0 \leq M \leq I$ with $\tau M \geq 1/2$, then*

$$\tau(MH) \geq \frac{1}{4} |H|_0^2 . \tag{3.3}$$

Proof. Set $\mu = \tau M$ and $t = \tau(MH)$. Scalar Jensen in an eigenbasis of H, using the diagonal entries of M and $I - M$ as weights, yields

$$\begin{aligned} \tau \log H &\leq \mu \log \frac{t}{\mu} + (1-\mu) \log \frac{1-t}{1-\mu} \\ &\leq \mu \log t + \log 2. \end{aligned}$$

This calculation does not require commutation of M and H. The endpoint $\mu = 1$ is interpreted by continuity. Since $\tau \log H \leq 0$ and $\mu \geq 1/2$, $\log t \geq 2\tau \log H - 2 \log 2$, proving the claim. □

Combining the two lemmas, every normalized positive input satisfies

$$\tau[M\Psi(G)] \geq \frac{1}{4} 2^{-2kA_m} |G|_2^{-4} \quad \text{if } \tau M \geq \tfrac{1}{2}. \tag{3.4}$$

Proposition 3.3 (Contraction over one block)**.** *For traceless Hermitian X, define*

$$V(X) = \max\left\{ |X|_1 , \frac{|X|_2}{2P_*} \right\}, \qquad \delta = 2^{-10-2kA_m} P_*^{-4}. \tag{3.5}$$

Then every block of m admissible steps obeys

$$V(\Psi X) \leq (1-\delta) V(X). \tag{3.6}$$

Proof. By homogeneity, take $V(X) = 1$, and write $x = |X|_1 \leq 1$. The zero case is immediate. Normalize the Jordan parts:

$$G_\pm = \frac{2X_\pm}{x}, \qquad \tau G_\pm = 1, \qquad |G_\pm|_2 \leq \frac{2\,|X|_2}{x} \leq \frac{4P_*}{x}.$$

For a binary test $M, I - M$, its sum of error probabilities is

$$e(M) = \tau[(I-M)\Psi G_+] + \tau[M\Psi G_-].$$

One of $M, I - M$ has normalized trace at least $1/2$. Equation (3.4) applied to the corresponding wrong state therefore gives

$$e(M) \geq \frac{1}{4} 2^{-2kA_m} \left(\frac{4P_*}{x} \right)^{-4} = \delta x^4.$$

The Helstrom identity yields

$$|\Psi X|_1 = x\big(1 - \min_M e(M)\big) \leq x - \delta x^5 \leq 1 - \delta.$$

The last inequality holds because $\delta \leq 2^{-10}$ and $x - \delta x^5$ increases on $[0, 1]$.

For the second norm, iterate (2.7) while using trace-norm contractivity at every intermediate step:

$$|\Psi X|_2 \le P_*^{1-\theta^m}\,|X|_1^{1-\theta^m}\,|X|_2^{\theta^m}\,.$$

Thus

$$\frac{|\Psi X|_2}{2P_*} \le 2^{-(1-\theta^m)}.$$

Since $\theta = 2q^2/(1+q^2) \le q$ and $q^{2m} \le 1/2$, $\theta^m \le 1/\sqrt{2}$. The contraction margin $1 - 2^{-(1-1/\sqrt{2})} > 0.18$ exceeds δ. Both components of V therefore contract by $1-\delta$. □

Proof of the upper bound in Theorem 1.1. For two echo states, $X = d(\rho - \rho')$ satisfies $V(X) \le 2$. If their histories share j blocks, Proposition 3.3 gives

$$D_{\mathrm{tr}}(\rho, \rho') \le (1-\delta)^j \le e^{-\delta j}.$$

Any remaining steps only decrease trace distance. Consequently,

$$W_{q,k}(\varepsilon) \le m \left\lceil \delta^{-1} \log \frac{1}{\varepsilon} \right\rceil.$$

The rate is

$$\delta^{-1} = 2^{10+2kA_m+4kq^2/(1-q^2)} \le 2^{10+6k\Lambda(q)},$$

because $A_m \le \Lambda(q)$ and the first term of $\Lambda(q)$ is $q^2/(1-q^2)$. This proves (1.4). □

4 A finite classical reservoir

Fix $\ell \ge 1$, let $M = 2^\ell$, and take an odd number N of data qubits with $N - 1 \ge \ell$. The complete reservoir will have $N + k$ qubits, with $k = \ell + 3$. For a data string $s \in \{0,1\}^N$, let

$$x_N(s) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} (-1)^{s_i}. \tag{4.1}$$

There is no zero magnetization. List the positive-magnetization strings $s_0, \ldots, s_{2^{N-1}-1}$ in decreasing magnetization, breaking ties arbitrarily. Pair each s_j with its bitwise complement $\bar{s}_j$.

Define three deterministic maps:

$$C_{\mathrm{h}}(s_j) = s_{\lfloor j/M \rfloor}, \qquad C_{\mathrm{h}}(\bar{s}_j) = \bar{s}_{\lfloor j/M \rfloor}, \tag{4.2}$$

$$C_{+}(s_j) = s_{\lfloor j/M \rfloor}, \qquad C_{+}(\bar{s}_j) = s_{\lfloor j/M \rfloor}, \tag{4.3}$$

$$C_{-}(s_j) = \bar{s}_{\lfloor j/M \rfloor}, \qquad C_{-}(\bar{s}_j) = \bar{s}_{\lfloor j/M \rfloor}. \tag{4.4}$$

The hold map moves each sign half toward its own extreme. The two loading maps also set the sign. The largest fiber has size M for hold and $2M$ for either load.

Proposition 4.1 (Implementation with a fixed permutation)**.** *The three maps are implemented by the model (1.1) with $k = \ell + 3$ refreshed qubits and a single fixed permutation unitary.*

Proof. Reserve $\ell+1$ refreshed ancillary qubits to retain the index within a fiber and two refreshed control qubits for the three modes. For each mode, the assignment

$$(s, 0^{\ell+1}, \text{mode}) \longmapsto (C_{\text{mode}}(s), \text{fiber index}, \text{mode})$$

is injective. The retained control values make different mode images disjoint. This injective assignment on a subset of the basis extends to a permutation of the entire basis. At every

step the input encoding prepares the chosen control value and the zero ancillary string. After the permutation, all data and ancillary qubits receive the prescribed local depolarization. The ancillary and control qubits are discarded at the next input reset. Their noisy values do not affect the retained data transition. □

On diagonal states, the data noise is independent bit flipping with probability $(1-q)/2$. Denote this strictly positive transition kernel by T_N. The hold data chain has kernel $P_N(s,t) = T_N(C_{\mathrm{h}}(s),t)$. It has a unique stationary law π_N, which is invariant under complementation.

All three modes induce the same exact Markov chain on complementary pairs $\{s,\bar{s}\}$. Indeed, their pre-noise outputs lie in the same pair, and T_N commutes with complementation. Write

$$e_N = \mathbb{P}_{\pi_N}\{\operatorname{sgn} x_N(S_{t+1}) \neq \operatorname{sgn} x_N(S_t)\} \tag{4.5}$$

for the stationary hold sign-flip probability.

Remark 4.2 (The exact state space). The finite magnetization by itself need not be a Markov chain: tied microstates can acquire slightly different output magnetizations under rank compression. The exact common Markov chain is the chain on complementary pairs. The limiting argument below works conditionally on each full microstate, uniformly over ties, and therefore does not assume finite magnetization lumpability.

5 Gaussian compression and its sign-flip bound

Let ϕ be the standard normal density and $Q(x) = \mathbb{P}(Z \geq x)$ its upper tail. For $x \geq 0$, put

$$F_\ell(x) = Q^{-1}\left(2^{-\ell} Q(x)\right). \tag{5.1}$$

This is the limit of dividing a Hamming-tail rank by 2^ℓ.

Lemma 5.1 (Growth under compression). *For every integer $\ell \geq 1$ and $x \geq 0$,*

$$x^2 + \frac{\ell}{4} \leq F_\ell(x)^2 \leq x^2 + 2\ell \log 2. \tag{5.2}$$

Proof. The Gaussian hazard $h(x) = \phi(x)/Q(x)$ satisfies $x \leq h(x) \leq x+1$ for $x \geq 0$. The lower bound follows from $h(x) = \mathbb{E}[Z \mid Z > x]$. For the upper bound,

$$h'(x) = 1 - \operatorname{Var}(Z \mid Z > x) \leq 1, \qquad h(0) < 1.$$

Writing $y = F_\ell(x)$, the quantile definition gives $\ell \log 2 = \int_x^y h(u)\,du$. If $y^2 - x^2 \leq \ell/4$, then

$$\ell \log 2 \leq \frac{y^2 - x^2}{2} + y - x \leq \frac{\ell}{8} + \frac{\sqrt{\ell}}{2} \leq \frac{5\ell}{8} < \ell \log 2,$$

a contradiction. Integrating $h(u) \geq u$ gives the upper bound. □

Consider the magnitude kernel on $[0,\infty)$

$$R' = |qF_\ell(R) + \sqrt{\eta} Z|, \qquad Z \sim N(0,1). \tag{5.3}$$

It is Feller, and Lemma 5.1 implies the drift bound

$$\mathbb{E}[(R')^2 \mid R = x] \leq q^2 x^2 + 2\ell q^2 \log 2 + \eta.$$

Cesaro averages of laws started at a finite point are therefore tight; any weak limit of such averages is invariant. Thus invariant laws exist. Uniqueness is unnecessary for the argument.

Proposition 5.2 (Stationary sign flips)**.** *For any invariant law ν of (5.3),*

$$e_\nu := \int Q\left(\frac{qF_\ell(x)}{\sqrt{\eta}}\right)\nu(dx) \le B(q,\ell) := \frac{\sqrt{\eta}}{2}\exp\left[-\frac{\ell}{8}\Lambda(q)\right]. \tag{5.4}$$

Proof. Set $a = \ell/4$ and $\mathcal{L}(s) = \int e^{-sx^2}\nu(dx)$. Stationarity, Gaussian integration, and $F_\ell(x)^2 \ge x^2 + a$ give

$$\mathcal{L}(s) \le (1+2\eta s)^{-1/2}e^{-as_1}\mathcal{L}(s_1), \qquad s_1 = \frac{q^2 s}{1+2\eta s}. \tag{5.5}$$

Starting at $s_0 = s$, the iterates and product are

$$s_j = \frac{sq^{2j}}{1+2s(1-q^{2j})}, \tag{5.6}$$

$$\prod_{j=0}^{J-1}(1+2\eta s_j) = 1+2s(1-q^{2J}). \tag{5.7}$$

Since $s_j \to 0$ and $\mathcal{L}(s_j) \to 1$, iteration gives

$$\mathcal{L}(s) \le (1+2s)^{-1/2}\exp\left[-a\sum_{j=1}^{\infty}s_j\right]. \tag{5.8}$$

The bound $Q(z) \le \frac{1}{2}e^{-z^2/2}$ for $z \ge 0$ follows by differentiating $e^{z^2/2}Q(z)$ and using $h(z) \ge z$. Take $s = q^2/(2\eta)$. Then

$$e_\nu \le \tfrac{1}{2}e^{-as}\mathcal{L}(s), \qquad s_j = \frac{q^{2(j+1)}}{2(1-q^{2(j+1)})}, \qquad (1+2s)^{-1/2} = \sqrt{\eta}.$$

Hence $s + \sum_{j\ge1} s_j = \frac{1}{2}\Lambda(q)$, and (5.4) follows. □

6 Passage to finite reservoirs and echo memory

We now justify that the Gaussian bound gives a witness inside the original finite-qubit model. Throughout this section, q and ℓ are fixed; the odd data size N tends to infinity.

6.1 Uniform approximation on compact magnetization intervals

If s_j has positive normalized magnetization x, then $j/2^N$ differs from the uniform-cube upper-tail probability at x by at most the mass of its Hamming-weight shell. Uniform binomial normal approximation and the vanishing maximum binomial atom imply

$$\frac{j}{2^N} = Q(x) + o(1) \tag{6.1}$$

uniformly for $0 \le x \le A$, including all positions within ties. The output rank is $\lfloor j/M \rfloor$; dividing by 2^N changes (6.1) to $Q(x)/M + o(1)$. On a fixed compact input interval this probability stays in a compact subinterval of $(0,1)$. Continuity of the inverse normal tail and binomial quantile convergence therefore give

$$|x_N(C_{\mathrm{h}}(s))| = F_\ell(|x_N(s)|) + o(1), \tag{6.2}$$

uniformly over all microstates with $|x_N(s)| \le A$. The same absolute output holds for either loading map.

Given any pre-noise data string with normalized magnetization y, its post-noise signed spins are independent. Each has variance η; the normalized sum has mean qy and variance η. For fixed q, a Berry–Esseen bound is $O_q(N^{-1/2})$, uniformly over the pre-noise string, because the summands are bounded. Combining this with (6.2) yields, for each bounded continuous test function g,

$$\mathbb{E}[g(|x_N(S')|) \mid S = s] = Kg(|x_N(s)|) + o(1), \tag{6.3}$$

uniformly on compact magnetization intervals, where

$$Kg(x) = \mathbb{E}\, g(|qF_\ell(x) + \sqrt{\eta}Z|).$$

For bounded continuous g, uniformity follows first for a compact output interval and then by Gaussian tail control. The conditional sign-flip probability similarly converges, uniformly on compact intervals, to

$$h_\ell(x) = Q\left(\frac{qF_\ell(x)}{\sqrt{\eta}}\right). \tag{6.4}$$

These statements concern conditional expectations given the full microstate; they do not assert an exact finite Markov property for magnetization.

6.2 Tightness of the stationary laws

Let μ_N be the law of $|x_N|$ under π_N. The full stationary reservoir is an all-hold echo state and has purity at most P_*. Taking the data marginal cannot increase the normalized Schatten 2-norm, so, writing $d_N = 2^N$,

$$|d_N \pi_N|_2 \le P_*.$$

For any data event A_N, Cauchy–Schwarz relative to the uniform cube therefore gives

$$\mathbb{P}_{\pi_N}(A_N) \le P_* \sqrt{\mathbb{P}_{\mathrm{unif}}(A_N)}.$$

Hoeffding's inequality for the uniform-cube magnetization yields

$$\mu_N((A,\infty)) \le \sqrt{2}\, P_* e^{-A^2/4}. \tag{6.5}$$

This proves tightness uniformly in N.

Choose a weakly convergent subsequence $\mu_{N_j} \Rightarrow \nu$. Stationarity of the full finite chain, (6.3), and (6.5) give

$$\int g\, d\nu = \int Kg\, d\nu$$

for bounded continuous g. Thus every subsequential limit is invariant for (5.3). Equations (6.4) and (6.5) also imply

$$e_{N_j} \longrightarrow \int h_\ell\, d\nu.$$

Proposition 5.2 applies to every such limit. Consequently,

$$\limsup_{N\to\infty} e_N \le B(q,\ell). \tag{6.6}$$

Because $B(q,\ell) > 0$, all sufficiently large odd finite N satisfy

$$e_N \le 2B(q,\ell) = \sqrt{\eta}\exp\left[-\frac{\ell}{8}\Lambda(q)\right]. \tag{6.7}$$

There is no requirement that the necessary size be uniform in q. The definition of $W_{q,k}$ optimizes over finite sizes separately for each parameter pair.

6.3 Two histories with a common suffix

Start from the all-hold echo state. Apply either load + or load −, then L identical hold inputs. These histories agree on their last L inputs.

The loading maps preserve the stationary distribution on complementary pairs. Load + gives the wrong decoded sign with probability e_N; load − has the same error probability. At every subsequent hold step, the sign-flip probability remains e_N, because it depends only on the complementary pair and that quotient distribution stays stationary. No independence of these errors is needed.

A union bound and measurement of the sign of x_N give

$$D_{\mathrm{tr}}(\rho_L^+, \rho_L^-) \geq 1 - 2(L+1)e_N. \tag{6.8}$$

The inequality remains valid for the full reservoir states, since the sign measurement acts on the data marginal.

Proof of the lower bound in Theorem 1.1. Take $\ell = k - 3 \geq 1$, choose a finite N satisfying (6.7), and put

$$A = \frac{1-\varepsilon}{4\sqrt{\eta}} \exp\left[\frac{\ell}{8}\Lambda(q)\right].$$

If $A < 1$, the asserted lower bound is zero and is immediate. Otherwise take $L = \lfloor A \rfloor - 1 \geq 0$. Then

$$2(L+1)e_N \leq \frac{1-\varepsilon}{2}, \qquad D_{\mathrm{tr}}(\rho_L^+, \rho_L^-) \geq \frac{1+\varepsilon}{2} > \varepsilon.$$

Forgetting fails at suffix length L; therefore $W_{q,k}(\varepsilon) \geq L + 1 = \lfloor A \rfloor$. This is (1.5). □

7 Comparison with the existing bounds

Put $p = 1 - q$. At fixed accuracy in the weak-noise limit, the current manuscript [1] and the refinement developed here give the following exponent scales:

	Current manuscript	Proposed refinement
Upper exponent	$O(k/p^2)$	$O((k/p)\log(1/p))$
Lower exponent	$\Omega(k\log(1/p))$	$\Omega((k/p)\log(1/p))$
Matching dependence	k, with noise fixed	k and weak-noise strength

The ratio of the old and new upper exponent scales is

$$\frac{k/p^2}{(k/p)\log(1/p)} = \frac{1}{p\log(1/p)} \longrightarrow \infty.$$

The new lower exponent is larger by a factor of order $1/p$. Thus the proposed result closes the weak-noise scaling gap as well as improving the upper bound.

For $k = 4$, (1.5) and (1.7) yield

$$\log W_{q,4}(\varepsilon) \geq \left(\frac{1}{16} + o(1)\right) \frac{\log(1/(1-q))}{1-q}.$$

Consequently $(1-q)\log W_{q,4}(\varepsilon)/4 \to \infty$, ruling out a uniform upper exponent $C_\varepsilon k/(1-q)$.

Scope. The result concerns arbitrary finite reservoir sizes, arbitrary global unitaries, fixed accuracy, and $k \geq 4$ for the matching lower bound. The upper proof permits all budgeted channels in (1.2). The lower construction uses exactly the reset-before-unitary architecture, and its readout is a bounded diagonal observable measuring the sign of the data magnetization. Its permutation can be very complex and its required size can be very large. The theorem characterizes optimal memory over the unrestricted model; it does not provide an efficient implementation. The constants in the exponent, optimal size, and exact accuracy dependence remain separate questions.

Status and scientific interpretation. This note records a proposed derivation for independent mathematical review. The key additional ingredients are the block estimate (3.2), the rank-compression construction, and the stationary finite-to-Gaussian transfer in Section 6. If validated, the result strengthens the fixed-noise law $\exp[\Theta_q(k)]$ into a matching law for fresh-qubit supply and noise strength. That is a substantive theoretical improvement; publication suitability is a separate assessment.

References

[1] M. Baumann, J. Stein, and C. Linnhoff-Popien, *The Fundamental Memory Ceiling of Noisy Quantum Reservoirs*, arXiv:2610.26228v1 (19 October 2026). arxiv.org/abs/2610.26228. The attached manuscript supplies the model, memory definition, and existing purity-cap argument (Supplement, Sec. III).

[2] T. Cubitt, M. Kastoryano, A. Montanaro, and K. Temme, *Quantum reverse hypercontractivity*, Journal of Mathematical Physics **56**, 102204 (2015). doi:10.1063/1.4933219. arXiv:1504.06143. The reverse-hypercontractive inequality used here is Theorem 4.

[3] C. King, *Hypercontractivity for semigroups of unital qubit channels*, Communications in Mathematical Physics **328**, 285 (2014). This is Ref. 33 of the attached manuscript and the source of its forward-hypercontractivity input.